Junxiu Lu

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Education

- **Tsinghua University** Beijing, China
  B.Eng. in Automation (Expected June 2027), GPA: 3.85/4.00 September 2023 - Present
  - **Selected Coursework:**
    Intelligent Computational Imaging, Signals and Systems Analysis, Linear Algebra, Probability and Statistics, Pattern Recognition and Machine Learning, Principles of Artificial Intelligence, and System Identification
  - **Research Interests:**
    Computational Imaging, Inverse Problem, Biomedical Imaging

Research Experience

- **Johns Hopkins University** Baltimore, MD, USA
  Visiting Student at Hopkins Computational Imaging Group July 2026 - Present
  Advisor: Yu Sun, Assitant Professor, Department of Electrical and Computer Engineering
  - **Project: Aberration-Aware Probabilistic Reconstruction for 3D Wide-Field Microscopy**
  - Investigating physics-informed generative methods for aberration-aware 3D wide-field fluorescence microscopy, with the goal of extending probabilistic reconstruction toward blind deconvolutions.
  - Reproduced the pretrained VOLT inference pipeline on simulated 3D fluorescence volumes and evaluated reconstructed outputs against ground-truth volumes.
  - Conducted comparative experiments on stochastic interpolants under ODE and SDE dynamics; visualized the temporal evolution of sample trajectories and probability densities.
- **Tsinghua University** Beijing, China
  Undergraduate Researcher at the Laboratory of Imaging and Intelligent Technology September 2025 - May 2026
  Advisor: Jiamin Wu, Associate Professor, Department of Automation
  - **Project: Computational Fundus Imaging System**
  - Assembled a prototype fundus imaging system, incorporating polarization control and background-frame subtraction to mitigate reflection artifacts in model-eye acquisitions.
  - Characterized the system's native optical resolution using a 1951 USAF target, resolving Group 7, Element 3 in raw measurements.
  - Reproduced a patch-based light-field reconstruction and computational aberration-correction algorithm, and applied image subtraction to suppress nonuniform glare.
- **Tsinghua University** Beijing, China
  Course Project of *Intelligent Computational Imaging* November 2025 - Janurary 2026
  Advisor: Jiamin Wu, Associate Professor, Department of Automation

- **Project: Calcium imaging of neuronal activity in the mouse cerebral cortex**
- Acquired light-field images of mouse cerebral cortex using the sLF microscopy system RUSH3D and implemented a computational imaging pipeline for 3D reconstruction.
- Rearranged raw light-field images, and applied a sliding-window stitching strategy to fuse angular views.
- Incorporated digital adaptive optics(DAO) into the pipeline to correct optical aberrations and DeepCAD video denoising model for deblurring.

Skills

- **Programming Languages:** C/C++, Python, MATLAB, Linux
- **Software & Tools:** MATLAB, PyCharm, VS Code, Visual Studio, LaTeX, Fiji ImageJ, Zemax

Languages

- **Mandarin Chinese:** Native
- **English:** Fluent
  - TOEFL(105/120): Reading 26, Listening 26, Speaking 26, Writing 27

Scholarships and Awards

- Tsinghua Friends - Zheng Geru Scholarship, Tsinghua University — Oct 2025
- Tsinghua Friends - Toyota Scholarship, Tsinghua University — Oct 2024